

HEALTHCARE FACILITY & PATIENT MANAGEMENT SYSTEM

A Detailed Data Analysis Report

Diabetes Dataset — Chapters 1 through 5

Prepared by

Akshintala Praneeth Kasyap

Student ID: 24162121024

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Dataset: Diabetes Dataset (diabetic_data.csv) — Kaggle

Repository: <https://github.com/praneeth-tech/HF-PMS>.git

Report Information

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1. Project Introduction and Environment Setup

Source document: Daily Report — 01/06/2026 | Week 2, Day 1

1.1 Overview and Objectives

*The primary objective of the first day of the internship was to establish the complete project environment for the **Healthcare Facility & Patient Management System** project. The focus was on setting up the required tools, understanding project requirements, organising project resources, configuring version control, and preparing the development environment for future data analysis tasks. The **Diabetes Dataset** was selected as the primary dataset, and all necessary platforms were configured to ensure smooth project execution throughout the internship.*

The specific objectives for this day were as follows:

- 1. Introduction to the Healthcare Facility & Patient Management System project.*
- 2. Understanding the project objectives and internship requirements.*
- 3. Familiarisation with the Diabetes Dataset from Kaggle.*
- 4. Setup of the project development environment.*
- 5. Configuration of GitHub, Git, Google Drive, Overleaf, and Google Colab.*
- 6. Establishment of project documentation and version control workflow.*
- 7. Creation of a structured project repository for future development.*
- 8. Preparation for Exploratory Data Analysis (EDA) activities.*
- 9. Organisation of project resources and datasets.*
- 10. Understanding the importance of synchronisation between local storage, GitHub, and Google Drive.*

1.2 Source Code — Section 1 (Google Colab)

The initial notebook (*01_Data_Loading_and_Understanding.ipynb*) was created in Google Colab. The following code blocks were written to begin the data loading phase.

1.2.1 Library Imports and Data Loading

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 df = pd.read_csv('/content/diabetic_data.csv')
```

Listing 1.1. Library imports and initial dataset load (Day 1)

1.2.2 Dataset Preview Functions

```
1 df.head()
2 df.tail()
3 df.sample(5)
4
5 df.shape
6
7 df.columns
8
9 df.info()
10 df.dtypes
```

Listing 1.2. Dataset preview — head, tail, sample, shape, columns (Day 1)

1.3 Source Code — Section 2 (GitHub Setup)

1.3.1 README.md Initialisation

The project *README.md* file was authored to document the project scope, dataset source, tooling, and planned analytical tasks. The content of the file is reproduced below.

```
# Healthcare Facility & Patient Management System
```


Project Overview

This project focuses on analyzing healthcare and patient data using the Diabetes Dataset. The objective is to understand patient health conditions, identify diabetes-related factors, and generate useful insights through Exploratory Data Analysis (EDA).

Dataset

Source: <https://www.kaggle.com/datasets/brandao/diabetes>

Tools Used

- Python*
- Pandas*
- NumPy*
- Matplotlib*
- Seaborn*
- Google Colab*
- GitHub*
- Google Drive*
- Overleaf*

Project Tasks

- Data Loading*
- Data Understanding*
- Data Cleaning*
- Exploratory Data Analysis (EDA)*
- Visualization*
- Documentation*

Author

Praneeth

1.3.2 Repository Folder Structure Creation (Command Prompt)

The project directory structure was established using the Windows Command Prompt. The following commands created the five principal project folders and uploaded the dataset to the repository.

```
1 mkdir data
2 mkdir notebooks
3 mkdir visualization
4 mkdir documentation
5 mkdir reports
6
7 git add .
8 git status
9 git commit -m "Created project structure"
10
11 git add .
12 git commit -m "Added diabetes dataset"
13 git push
```

Listing 1.3. Repository folder creation and Git initialisation (Day 1)

1.4 Work Completed

1.4.1 Project Initialisation

- Reviewed internship guidelines and project requirements.
- Understood the Healthcare Facility & Patient Management System project scope.
- Identified the Diabetes Dataset as the primary dataset for analysis.

1.4.2 Environment Setup

- Created and configured GitHub account and repository (HF-PMS).
- Created Google Drive workspace for project storage and backup.
- Created Overleaf project for documentation and report preparation.
- Verified Google Colab accessibility for future notebook development.

1.4.3 Git and Version Control Setup

- *Installed Git on the local machine.*
- *Configured Git user credentials.*
- *Verified successful Git installation.*
- *Learned basic Git workflow and version control concepts.*

1.4.4 Repository Configuration

- *Cloned the HF-PMS repository to the local development environment.*
- *Verified repository synchronisation between local system and GitHub.*
- *Tested repository connectivity using Git commands.*

1.4.5 Project Structure Creation

The following project folders were created:

- *data/ — primary dataset storage.*
- *notebooks/ — Jupyter notebooks for each analytical phase.*
- *reports/ — daily and weekly progress reports.*
- *visualizations/ — generated visualisation outputs.*
- *documentation/ — reference materials and documentation.*

Tracking files (.gitkeep) were added to each folder to allow Git management of empty directories.

1.4.6 Dataset Management

- *Downloaded the Diabetes Dataset from Kaggle.*
- *Extracted dataset files from the downloaded archive.*
- *Identified the primary CSV file (diabetic_data.csv).*
- *Stored the dataset in the data/ directory.*

1.4.7 Git Operations Performed

- *Executed `git status` to verify repository state.*
- *Added project files using `git add`.*
- *Committed changes with `git commit -m "Created project structure"`.*

- *Pushed updates to GitHub using `git push`.*
- *Successfully synchronised local and remote repositories.*

1.4.8 Documentation Activities

- *Created the initial `README.md` file.*
- *Prepared the reporting structure for daily and weekly reports.*
- *Established documentation standards for the internship project.*

1.5 Summary

Day 1 focused on establishing the foundation for the Healthcare Facility & Patient Management System project. The project requirements and internship workflow were reviewed to understand the expected deliverables and development process. Essential platforms including GitHub, Git, Google Drive, Overleaf, and Google Colab were configured to support project development, documentation, and collaboration activities.

A GitHub repository named HF-PMS was successfully created and connected to the local development environment using Git. A structured project architecture was implemented by creating dedicated folders for datasets, notebooks, reports, visualisations, and documentation. The Diabetes Dataset was downloaded from Kaggle, extracted, and organised within the project repository for future analysis.

Version control practices were introduced and implemented through Git operations such as repository cloning, file tracking, committing changes, and pushing updates to GitHub. The successful synchronisation between the local machine and GitHub confirmed that the project environment was fully operational. By the end of the day, all setup activities were completed successfully, and the project was prepared for the next phase involving dataset exploration and analysis.

1.6 Outcomes Achieved

- *Development environment successfully configured.*
- *Project repository fully operational.*
- *Dataset acquired and organised.*
- *Version control workflow established.*

- *Project structure prepared for Exploratory Data Analysis (EDA).*
- *GitHub and local repository successfully synchronised.*

2. Dataset Loading and Understanding

Source document: Daily Report — 02/06/2026 | Week 2, Day 2

2.1 Overview and Objectives

The objective of Day 2 was to gain a comprehensive understanding of the Diabetes Dataset before performing any analytical operations. The dataset was loaded into Google Colab and examined to understand its structure, dimensions, attributes, and data types. Basic statistical summaries were generated to evaluate the characteristics of the data, while feature analysis helped identify numerical and categorical variables. This activity served as a critical foundation for future data cleaning, visualisation, and exploratory data analysis tasks.

The specific key points addressed on this day were:

- Load the Diabetes Dataset into Google Colab.
- Verify successful dataset loading.
- Understand the dataset structure and dimensions.
- Review column names and feature descriptions.
- Analyse data types of all attributes.
- Generate dataset information and summary statistics.
- Identify numerical and categorical features.
- Create a feature summary table for documentation.
- Understand the overall quality and characteristics of the dataset.
- Prepare the dataset for future data cleaning and exploratory data analysis.

2.2 Source Code — Section 1 (Google Colab)

2.2.1 Step 1 — Data Loading

```
1 import pandas as pd
```

```
2
3 df = pd.read_csv('diabetic_data.csv')
```

Listing 2.1. Dataset loading with Pandas (Day 2, Step 1)

2.2.2 Step 2 — Dataset Preview

```
1 df.head()
2 df.tail()
3 df.sample(5)
```

Listing 2.2. Dataset preview functions (Day 2, Step 2)

2.2.3 Step 3 — Shape

```
1 df.shape
```

Listing 2.3. Dataset shape (Day 2, Step 3)

Output: (101766, 50) — the dataset contains **101,766 rows and 50 columns**.

2.2.4 Step 4 — Column Names

```
1 df.columns.tolist()
```

Listing 2.4. Column name extraction (Day 2, Step 4)

The complete list of 50 column names returned was:

```
['encounter_id', 'patient_nbr', 'race', 'gender', 'age', 'weight',
 'admission_type_id', 'discharge_disposition_id', 'admission_source_id',
 'time_in_hospital', 'payer_code', 'medical_specialty',
 'num_lab_procedures', 'num_procedures', 'num_medications',
 'number_outpatient', 'number_emergency', 'number_inpatient',
 'diag_1', 'diag_2', 'diag_3', 'number_diagnoses', 'max_glu_serum',
 'A1Cresult', 'metformin', 'repaglinide', 'nateglinide',
 'chlorpropamide', 'glimepiride', 'acetoheexamide', 'glipizide',
 'glyburide', 'tolbutamide', 'pioglitazone', 'rosiglitazone',
 'acarbose', 'miglitol', 'troglitazone', 'tolazamide', 'examide',
 'citoglipton', 'insulin', 'glyburide-metformin', 'glipizide-metformin',
```

```
'glimepiride-pioglitazone', 'metformin-rosiglitazone',  
'metformin-pioglitazone', 'change', 'diabetesMed', 'readmitted']
```

2.2.5 Step 5 — Dataset Information

```
1 df.info()
```

Listing 2.5. Dataset info (Day 2, Step 5)

Output summary:

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 101766 entries, 0 to 101765  
Data columns (total 50 columns):  
#      Column                                Non-Null Count  Dtype  
---  -  
0     encounter_id                            101766 non-null  int64  
1     patient_nbr                             101766 non-null  int64  
2     race                                    101766 non-null  object  
3     gender                                  101766 non-null  object  
4     age                                     101766 non-null  object  
5     weight                                 101766 non-null  object  
6     admission_type_id                      101766 non-null  int64  
7     discharge_disposition_id              101766 non-null  int64  
8     admission_source_id                   101766 non-null  int64  
9     time_in_hospital                      101766 non-null  int64  
10    payer_code                             101766 non-null  object  
11    medical_specialty                     101766 non-null  object  
12    num_lab_procedures                    101766 non-null  int64  
13    num_procedures                        101766 non-null  int64  
14    num_medications                       101766 non-null  int64  
15    number_outpatient                      101766 non-null  int64  
16    number_emergency                      101766 non-null  int64  
17    number_inpatient                      101766 non-null  int64  
18    diag_1                                101766 non-null  object  
19    diag_2                                101766 non-null  object  
20    diag_3                                101766 non-null  object
```



```
21  number_diagnoses      101766 non-null  int64
22  max_glu_serum         5346 non-null   object
23  A1Cresult             17018 non-null  object
24  metformin             101766 non-null  object
    ... (columns 25-49: medication/outcome columns, all 101766 non-null)
dtypes: int64(13), object(37)
memory usage: 38.8+ MB
```

2.2.6 Step 6 — Data Types

```
1 df.dtypes
```

Listing 2.6. Column data types (Day 2, Step 6)

The dataset contains **13 integer (int64) columns** and **37 object (string/categorical) columns**, confirming the mixed feature composition of the dataset.

2.2.7 Step 7 — Descriptive Statistics (Numerical)

```
1 df.describe()
```

Listing 2.7. Numerical descriptive statistics (Day 2, Step 7)

Table 2.1. Selected Descriptive Statistics for Key Numerical Features (Day 2)

Feature	Mean	Std	Min	50%	Max
time_in_hospital	4.40	2.99	1	4	14
num_lab_procedures	43.11	19.10	1	44	132
num_procedures	1.34	1.71	0	1	6
num_medications	15.30	8.13	1	14	81
number_outpatient	0.37	0.88	0	0	42
number_emergency	0.20	0.93	0	0	76
number_inpatient	0.64	1.26	0	0	21
number_diagnoses	7.42	1.93	1	8	16

2.2.8 Step 8 — Extended Descriptive Statistics (All Columns)

```
1 df.describe(include='all')
```

Listing 2.8. Extended statistics including categorical columns (Day 2, Step 8)

Key observations from the extended statistics:

- **race:** 6 unique values; most frequent value is Caucasian (76,099 occurrences).
- **gender:** 3 unique values; most frequent is Female (54,708 occurrences).
- **age:** 10 age bracket values; most frequent bracket is [70-80) with 26,068 occurrences.
- **weight:** 10 unique values; most frequent is ? (98,569 occurrences, indicating systemic non-recording).
- **readmitted:** 3 categories — NO, >30, <30.

2.2.9 Step 9 — Unique Value Counts per Column

```
1 for col in df.columns:
2     print(col, df[col].nunique())
```

Listing 2.9. Unique value enumeration per column (Day 2, Step 9)

Table 2.2. Unique Value Counts for Selected Features (Day 2)

Feature	Unique	Feature	Unique
encounter_id	101,766	diag_1	717
patient_nbr	71,518	diag_2	749
race	6	diag_3	790
gender	3	max_glu_serum	3
age	10	A1Cresult	3
weight	10	metformin	4
admission_type_id	8	insulin	4
discharge_disposition_id	26	change	2
admission_source_id	17	diabetesMed	2
time_in_hospital	14	readmitted	3
medical_specialty	73	num_lab_procedures	118
num_procedures	7	num_medications	75
number_outpatient	39	number_emergency	33
number_inpatient	21	number_diagnoses	16

2.2.10 Step 10 — Feature Summary Table

```
1 summary = pd.DataFrame({
2     'Column'      : df.columns,
3     'DataType'    : df.dtypes.values,
4     'MissingValues': df.isnull().sum().values,
5     'UniqueValues': df.nunique().values
6 })
7 summary
```

Listing 2.10. Feature summary table generation (Day 2, Step 10)

This table confirmed that, in the raw dataset as initially loaded, only `max_glu_serum` (96,420 missing) and `A1Cresult` (84,748 missing) contained significant missing value counts. All remaining 48 features had zero or negligible missing values at this stage.

2.3 Source Code — Section 2 (GitHub Operations)

At the end of Day 2, the completed notebook was committed and pushed to the remote repository.

```
1 git add .
2 git commit -m "Added dataset understanding notebook"
3 git push
```

Listing 2.11. Git add, commit, and push operations (Day 2)

Repository: <https://github.com/praneeth-tech/HF-PMS.git>

2.4 Work Completed

2.4.1 Dataset Loading

- Loaded the Diabetes Dataset into Google Colab.
- Verified successful dataset import using Pandas.
- Confirmed dataset accessibility for analysis.

2.4.2 Dataset Exploration

- Examined the first and last records using `head()` and `tail()`.

- Reviewed dataset dimensions: 101,766 rows $\times$ 50 columns.
- Inspected column names and available features.
- Analysed the overall dataset structure.

2.4.3 Data Understanding

- Identified data types: 13 integer columns and 37 object columns.
- Reviewed non-null value counts for each column.
- Generated dataset information using `df.info()`.
- Studied the characteristics of healthcare-related features.

2.4.4 Statistical Analysis

- Generated summary statistics for numerical features using `df.describe()`.
- Reviewed count, mean, standard deviation, minimum, and maximum values.
- Evaluated the distribution of key numerical attributes.

2.4.5 Feature Analysis

- Identified 13 numerical and 37 categorical features.
- Analysed unique value cardinality across all 50 columns.
- Created a feature summary table for documentation purposes.

2.4.6 Version Control Activities

- Added the completed notebook to Git tracking.
- Created a project commit for dataset understanding activities.
- Successfully pushed updates to the GitHub repository.

2.5 Summary

Today's work focused on understanding the Diabetes Dataset and preparing it for future analytical activities. The dataset was successfully loaded into Google Colab, and its structure, dimensions, and features were examined in detail. Various dataset exploration techniques were performed to gain insights into the available attributes, data types, and overall composition of the data.

Basic statistical analysis was conducted using summary statistics, and feature-level analysis

was performed to identify numerical and categorical variables. Dataset information, unique values, and attribute characteristics were reviewed to develop a clear understanding of the healthcare-related data available for analysis. The completed work was documented in the notebook *01_Data_Loading_and_Understanding.ipynb* and synchronised with the GitHub repository using Git version control.

By the end of the session, a comprehensive understanding of the dataset structure and feature composition was achieved, providing a strong foundation for the next phase involving missing value analysis, data quality assessment, and exploratory data analysis (EDA).

2.6 Outcomes Achieved

- *Successfully understood the structure of the Diabetes Dataset.*
- *Documented dataset characteristics and feature information.*
- *Prepared the dataset for missing value analysis and future EDA activities.*
- *Completed the Dataset Understanding phase of the project.*

3. Data Quality Assessment

Source document: Daily Report — 03/06/2026 | Week 2, Day 3

3.1 Overview and Objectives

The objective of Day 3 was to evaluate the quality and reliability of the Diabetes Dataset before proceeding with advanced exploratory analysis. The dataset was examined for missing values, duplicate records, and feature completeness to identify potential issues that may impact analysis results. Data quality reports and visualisations were generated to better understand the condition of the dataset.

The specific key points addressed were:

- *Assess the overall quality of the Diabetes Dataset.*
- *Identify missing values and incomplete records.*
- *Calculate missing value counts and percentages.*
- *Detect duplicate records within the dataset.*
- *Evaluate feature completeness and reliability.*
- *Classify numerical and categorical features.*
- *Generate data quality reports and visualisations.*
- *Prepare the dataset for Exploratory Data Analysis (EDA).*
- *Identify potential data issues that may affect future analysis.*
- *Document observations and findings for further processing.*

3.2 Source Code — Section 1 (Google Colab)

3.2.1 Step 1 — Library Imports

```
1 import pandas as pd
2 import numpy as np
```

```
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 pd.set_option('display.max_columns', None)
7 pd.set_option('display.max_rows', 100)
```

Listing 3.1. Library imports with display options (Day 3, Step 1)

3.2.2 Step 2 — Data Loading

```
1 df = pd.read_csv('/content/diabetic_data.csv')
```

Listing 3.2. Dataset loading (Day 3, Step 2)

3.2.3 Step 3 — Head and Shape

```
1 df.head()
2 df.shape
```

Listing 3.3. Head and shape inspection (Day 3, Step 3)

Output: (100227, 50)

*Note: The shape at this stage returned **100,227 rows** rather than 101,766. This discrepancy reflects that filtering or pre-processing had already been applied to the dataset version used in this session.*

3.2.4 Step 4 — Null Value Detection

```
1 df.isnull().sum()
```

Listing 3.4. Null value count per column (Day 3, Step 4)

Selected output (features with missing values):

```
max_glu_serum    94899
A1Cresult        83483
(all other columns: 1 each, or 0)
```

3.2.5 Step 5 — Missing Values Percentage Analysis

```
1 missing = df.isnull().sum()
```

```
2 missing_df = pd.DataFrame({
3     'Missing Count'      : missing,
4     'Missing Percentage': round((missing / len(df)) * 100, 2)
5 })
6 missing_df.sort_values(by='Missing Count', ascending=False)
```

Listing 3.5. Missing value count and percentage table (Day 3, Step 5)

Table 3.1. Top Missing Value Results — Sorted by Missing Count (Day 3)

PFeature	PMissing Count	PMissing %
max_glu_serum	94,899	94.68%
A1Cresult	83,483	83.29%
payer_code	1	0.00%
medical_specialty	1	0.00%
age	1	0.00%
weight	1	0.00%
admission_type_id	1	0.00%
discharge_disposition_id	1	0.00%
admission_source_id	1	0.00%
time_in_hospital	1	0.00%
All remaining features	1 each	0.00%

The analysis confirms that **max_glu_serum** (94.68% missing) and **A1Cresult** (83.29% missing) are the only features with analytically significant rates of missing data.

3.2.6 Step 6 — Missing Values Heatmap

```
1 plt.figure(figsize=(12, 6))
2 sns.heatmap(
3     df.isnull(),
4     cbar=False,
5     yticklabels=False
6 )
7 plt.title('Missing Values Heatmap')
```



```
8 plt.show()
```

Listing 3.6. Missing values heatmap generation (Day 3, Step 6)

The heatmap visually confirmed the pattern identified in Step 5: missing values appear as two narrow vertical bands corresponding exclusively to the `max_glu_serum` and `A1Cresult` columns. All other columns appear uniformly dark (no missing values), demonstrating strong overall data completeness.

3.2.7 Step 7 — Duplicate Record Detection

```
1 duplicates = df.duplicated().sum()
2 print("Duplicate Records:", duplicates)
```

Listing 3.7. Duplicate record count (Day 3, Step 7)

Output: Duplicate Records: 0

No duplicate records were detected in the dataset.

3.2.8 Step 8 — Displaying Duplicate Rows

```
1 df[df.duplicated()]
```

Listing 3.8. Display of duplicate rows — verification (Day 3, Step 8)

The result confirmed an empty DataFrame, validating the absence of any duplicate encounter records.

3.2.9 Step 9 — Feature Completeness Analysis

```
1 feature_quality = pd.DataFrame({
2     'Column'      : df.columns,
3     'Data Type'    : df.dtypes.values,
4     'Missing Values': df.isnull().sum().values,
5     'Unique Values': df.nunique().values
6 })
7 feature_quality
```

Listing 3.9. Feature quality summary table (Day 3, Step 9)

Table 3.2. Feature Quality Summary — Selected Rows (Day 3)

P#	PColumn	PData Type	PMissing	PUnique
0	<i>encounter_id</i>	int64	0	100,227
1	<i>patient_nbr</i>	int64	0	70,642
2	<i>race</i>	object	1	6
3	<i>gender</i>	object	1	3
4	<i>age</i>	object	1	10
5	<i>weight</i>	object	1	10
6	<i>admission_type_id</i>	float64	1	8
7	<i>discharge_disposition_id</i>	float64	1	26
8	<i>admission_source_id</i>	float64	1	17
9	<i>time_in_hospital</i>	float64	1	14
10	<i>payer_code</i>	object	1	18
11	<i>medical_specialty</i>	object	1	73
12	<i>num_lab_procedures</i>	float64	1	118
13	<i>num_procedures</i>	float64	1	7
14	<i>num_medications</i>	float64	1	75
15	<i>number_outpatient</i>	float64	1	39
16	<i>number_emergency</i>	float64	1	32
17	<i>number_inpatient</i>	float64	1	21
18	<i>diag_1</i>	object	1	716
19	<i>diag_2</i>	object	1	747
20	<i>diag_3</i>	object	1	786
21	<i>number_diagnoses</i>	float64	1	16
22	<i>max_glu_serum</i>	object	94,899	3
23	<i>A1Cresult</i>	object	83,483	3

3.2.10 Step 10 — Feature Classification

```

1 numeric_cols = df.select_dtypes(include=np.number).columns
2 categorical_cols = df.select_dtypes(include=['object', '
    category']).columns
3
4 print("Numerical Features:")

```

```

5 print(list(numeric_cols))
6 print("\nCategorical Features:")
7 print(list(categorical_cols))

```

Listing 3.10. Numerical and categorical feature identification (Day 3, Step 10)

Numerical Features (13):

```

['encounter_id', 'patient_nbr', 'admission_type_id',
 'discharge_disposition_id', 'admission_source_id',
 'time_in_hospital', 'num_lab_procedures', 'num_procedures',
 'num_medications', 'number_outpatient', 'number_emergency',
 'number_inpatient', 'number_diagnoses']

```

Categorical Features (37):

```

['race', 'gender', 'age', 'weight', 'payer_code',
 'medical_specialty', 'diag_1', 'diag_2', 'diag_3',
 'max_glu_serum', 'A1Cresult', 'metformin', 'repaglinide',
 'nateglinide', 'chlorpropamide', 'glimepiride', 'acetoheaxamide',
 'glipizide', 'glyburide', 'tolbutamide', 'pioglitazone',
 'rosiglitazone', 'acarbose', 'miglitol', 'troglitazone',
 'tolazamide', 'examide', 'citoglipton', 'insulin',
 'glyburide-metformin', 'glipizide-metformin',
 'glimepiride-pioglitazone', 'metformin-rosiglitazone',
 'metformin-pioglitazone', 'change', 'diabetesMed', 'readmitted']

```

3.2.11 Step 11 — Data Quality Summary

```

1 print("Total Rows      :", df.shape[0])
2 print("Total Columns :", df.shape[1])
3 print("Total Missing Values :", df.isnull().sum().sum())
4 print("Duplicate Records      :", df.duplicated().sum())

```

Listing 3.11. Consolidated data quality summary output (Day 3, Step 11)

Output:

```
Total Rows      : 100227
```

Total Columns	: 50
Total Missing Values	: 178428
Duplicate Records	: 0

3.2.12 Step 12 — Saving the Data Quality Report

```
1 feature_quality.to_csv('Data_Quality_Report.csv', index=False  
    )
```

Listing 3.12. Export feature quality report as CSV (Day 3, Step 12)

The CSV report was saved locally and backed up to Google Drive.

3.3 Source Code — Section 2 (GitHub Operations)

```
1 git add .  
2 git commit -m "Completed data quality assessment"  
3 git push
```

Listing 3.13. Git operations for Day 3 notebook commit (Day 3)

3.4 Work Completed

3.4.1 Data Quality Assessment

- Performed comprehensive data quality evaluation of the Diabetes Dataset.
- Verified dataset integrity and readiness for further analysis.
- Reviewed dataset consistency and completeness.

3.4.2 Missing Value Analysis

- Identified missing values across all dataset features.
- Calculated missing value counts for each column.
- Computed missing value percentages.
- Generated a missing value summary table.
- Analysed the impact of missing data on dataset quality.

3.4.3 Data Visualisation

- Created a Missing Values Heatmap using Seaborn.
- Visualised the distribution of missing data across all features.
- Identified patterns of missing values concentrated in `max_glu_serum` and `A1Cresult`.
- Evaluated feature completeness through graphical representation.

3.4.4 Duplicate Record Analysis

- Checked the dataset for duplicate entries.
- Confirmed zero duplicate records.
- Assessed the integrity of the encounter-level data collection.

3.4.5 Feature Quality Assessment

- Generated a feature quality summary table.
- Reviewed data types of all 50 features.
- Analysed unique value cardinality for each attribute.
- Evaluated feature-level completeness and reliability.

3.4.6 Feature Classification

- Identified 13 numerical features within the dataset.
- Identified 37 categorical features within the dataset.
- Categorised dataset attributes for future analytical tasks.
- Prepared feature groups for Exploratory Data Analysis (EDA).

3.4.7 Documentation and Reporting

- Created the notebook `02_Data_Quality_Assessment.ipynb`.
- Documented data quality findings and observations.
- Generated a CSV data quality report for future reference.
- Organised outputs within the project structure.

3.5 Summary

Today's work focused on assessing the quality and reliability of the Diabetes Dataset before proceeding to advanced exploratory analysis. A comprehensive data quality assessment was performed to identify missing values, evaluate feature completeness, and examine the overall condition of the dataset. Missing value counts and percentages were calculated for all attributes, and a missing value heatmap was generated to visualise data completeness and identify potential patterns of missing information.

In addition to missing value analysis, the dataset was examined for duplicate records and feature-level quality metrics. Numerical and categorical features were identified and categorised to support future analytical activities. A feature quality summary table was created to provide a consolidated view of data types, unique values, and missing value information for each attribute.

All findings were documented within the notebook `02_Data_Quality_Assessment.ipynb`, and the completed work was synchronised with the GitHub repository using Git version control. By the end of the session, a clear understanding of the dataset's quality, completeness, and reliability was achieved, establishing a strong foundation for the upcoming Exploratory Data Analysis (EDA) and visualisation phase.

3.6 Outcomes Achieved

- *Successfully evaluated the quality of the Diabetes Dataset.*
- *Identified missing value patterns and data completeness.*
- *Assessed duplicate records and confirmed zero duplicates.*
- *Prepared the dataset for the Exploratory Data Analysis phase.*
- *Established a clear understanding of dataset quality before advanced analysis.*

4. Exploratory Data Analysis (EDA)

Source document: Daily Report — 04/06/2026 | Week 2, Day 4

4.1 Overview and Objectives

The objective of Day 4 was to perform Exploratory Data Analysis (EDA) on the Diabetes Dataset to gain deeper insights into the data. Various statistical and visualisation techniques were used to understand feature distributions, identify trends, detect outliers, and examine relationships between variables. Visualisations such as histograms, KDE plots, boxplots, correlation heatmaps, and pair plots were generated to reveal hidden patterns within the dataset.

The specific key points addressed were:

- Perform Exploratory Data Analysis (EDA) on the Diabetes Dataset.*
- Analyse the distribution of numerical features.*
- Generate statistical summaries for better understanding.*
- Create histograms to visualise data distribution.*
- Generate KDE plots to study data density and trends.*
- Detect outliers using boxplot visualisations.*
- Analyse relationships between different features.*
- Create a correlation matrix and heatmap.*
- Generate pair plots for feature interaction analysis.*
- Identify significant patterns and healthcare-related insights.*
- Evaluate feature skewness and data distribution characteristics.*
- Prepare findings for reporting and future predictive modelling.*

4.2 Source Code — Section 1 (Google Colab)

4.2.1 Step 1 — Library Imports

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 pd.set_option('display.max_columns', None)
7 pd.set_option('display.max_rows', 100)
```

Listing 4.1. Library imports with display options (Day 4, Step 1)

4.2.2 Step 2 — Data Loading

```
1 df = pd.read_csv('/content/diabetic_data.csv')
```

Listing 4.2. Dataset loading (Day 4, Step 2)

4.2.3 Step 3 — Descriptive Statistics

```
1 df.describe()
```

Listing 4.3. Statistical summary using describe() (Day 4, Step 3)

Selected output (partial, key numerical features):

Table 4.1. Descriptive Statistics Output — Key Columns (Day 4)

Statistic	time_in	num_lab	num	num	num	num
	Phospital	Pproc.	Pproc.	Pmeds	Pemerg.	Pinpat.
count	44,110	44,110	44,110	44,110	44,110	44,110
mean	4.58	43.19	1.39	15.30	0.12	0.59
std	3.10	19.10	1.70	8.13	0.62	1.21
min	1	1	0	1	0	0
25%	2	31	0	10	0	0
50%	4	44	1	14	0	0
75%	7	57	2	20	0	1
max	14	129	6	81	42	21

4.2.4 Step 4 — Histograms

```
1 df.hist(figsize=(18, 12), bins=20)
2 plt.tight_layout()
3 plt.show()
```

Listing 4.4. Histogram generation for all numerical features (Day 4, Step 4)

Key distributional observations from the histograms:

- *encounter_id* and *patient_nbr*: approximately uniform distributions, consistent with identifier fields.
- *time_in_hospital*: right-skewed; highest frequency in the 2–4 day range, tapering progressively at longer durations.
- *num_lab_procedures*: near-normal distribution centred around 40–50 procedures.
- *num_medications*: right-skewed; peak in the 10–20 medication range.
- *number_outpatient*, *number_emergency*, *number_inpatient*: extreme right-skew, with the vast majority of values concentrated at or near zero.
- *number_diagnoses*: multi-modal distribution; distinct peaks particularly at values 7–9.

4.2.5 Step 5 — KDE Distribution Plots

```
1 numeric_cols = df.select_dtypes(include=np.number).columns
2 for col in numeric_cols:
3     plt.figure(figsize=(8, 4))
4     sns.kdeplot(df[col])
5     plt.title(col)
6     plt.show()
```

Listing 4.5. KDE plots for all numerical features (Day 4, Step 5)

The KDE plots reinforced and extended histogram observations. *number_emergency* and *number_outpatient* showed sharp, spike-like density peaks at zero, confirming extreme zero-inflation. *num_medications* revealed a clean unimodal distribution with a right tail extending above 40. *time_in_hospital* displayed a characteristic right-skewed wave pattern with multiple local peaks at each integer day value, reflecting its discrete nature.

4.2.6 Step 6 — Boxplots (Outlier Detection)

```
1 for col in numeric_cols:
2     plt.figure(figsize=(8, 4))
3     sns.boxplot(x=df[col])
4     plt.title(col)
5     plt.show()
```

Listing 4.6. Boxplot generation per numerical feature (Day 4, Step 6)

Outlier observations from boxplots:

- *num_lab_procedures*: dense cluster of outlier points above the upper whisker, concentrated in the 90–132 range.
- *num_medications*: extensive outlier range from approximately 32 to 81 medications.
- *number_outpatient*: outlier markers extending to 35.
- *number_emergency*: outlier markers extending to 42.
- *number_inpatient*: outlier markers extending to 21.
- *discharge_disposition_id*: numerous discrete outliers above value 11.
- *time_in_hospital*: two visible outliers at 13 and 14 days.
- *admission_type_id*: one outlier at value 8.

4.2.7 Step 7 — Correlation Matrix

```
1 corr = df[numeric_cols].corr()
2 corr
```

Listing 4.7. Pearson correlation matrix computation (Day 4, Step 7)

4.2.8 Step 8 — Correlation Heatmap

```
1 plt.figure(figsize=(14, 10))
2 sns.heatmap(
3     corr,
4     annot=True,
5     cmap='coolwarm',
6 )
7 plt.title('Correlation Matrix')
8 plt.show()
```

Listing 4.8. Correlation heatmap visualisation (Day 4, Step 8)

Notable correlation values extracted from the heatmap output:

Table 4.2. Notable Pairwise Correlation Coefficients (Day 4)

PFeature Pair	PCorrelation	PStrength
num_procedures ↔ num_medications	0.4788	Moderate–Strong
num_medications ↔ time_in_hospital	0.4775	Moderate
admission_type_id ↔ admission_source_id	0.3656	Moderate
time_in_hospital ↔ num_lab_procedures	0.3235	Weak–Moderate
number_emergency ↔ number_inpatient	0.2225	Weak
number_diagnoses ↔ time_in_hospital	0.2660	Weak
number_diagnoses ↔ num_lab_procedures	0.1763	Weak

4.2.9 Step 9 — Pair Plot

```
1 sns.pairplot(df[numeric_cols])
2 plt.show()
```

Listing 4.9. Pair plot for all numerical features (Day 4, Step 9)

The 13×13 pair plot grid confirmed the structural observations from preceding analyses. The most visually coherent scatter relationships appear between num_procedures and num_medications, and between num_medications and time_in_hospital. Identifier columns (encounter_id, patient_nbr) displayed cloud-like scatter patterns, consistent with their role as non-informative index fields.

4.2.10 Step 10 — Skewness Analysis

```
1 df[numeric_cols].skew()
```

Listing 4.10. Skewness coefficient computation (Day 4, Step 10)

Output:

Table 4.3. Feature Skewness Coefficients (Day 4)

PFeature	PSkewness	PClassification
number_emergency	18.326009	Heavily Right-Skewed
number_outpatient	9.536715	Heavily Right-Skewed
number_inpatient	3.660828	Right-Skewed
discharge_disposition_id	2.013653	Right-Skewed
num_medications	1.485678	Right-Skewed
admission_type_id	1.264269	Right-Skewed
num_procedures	1.230869	Right-Skewed
patient_nbr	1.023544	Right-Skewed
admission_source_id	0.984582	Right-Skewed
time_in_hospital	1.061432	Right-Skewed
num_lab_procedures	-0.123162	Approximately Normal
encounter_id	-0.146673	Approximately Normal
number_diagnoses	-0.473474	Slightly Left-Skewed

4.3 Source Code — Section 2 (GitHub Operations)

```
1 git add .
2 git commit -m "Completed exploratory data analysis and
   visualizations"
3 git push
```

Listing 4.11. Git commit for EDA notebook (Day 4)

4.4 Work Completed

4.4.1 Exploratory Data Analysis

- Performed EDA on the Diabetes Dataset.
- Examined overall data distribution and feature characteristics.
- Generated statistical summaries for numerical attributes.

- *Analysed key healthcare-related variables.*

4.4.2 Distribution Analysis

- *Created histograms to visualise the distribution of numerical features.*
- *Generated KDE plots to study data density and distribution patterns.*
- *Identified skewed and normally distributed attributes.*

4.4.3 Outlier Detection

- *Created boxplots for all 13 numerical features.*
- *Identified potential outliers and extreme observations.*
- *Analysed the impact of outliers on dataset quality.*

4.4.4 Correlation Analysis

- *Calculated Pearson correlations between all numerical features.*
- *Generated a full correlation matrix.*
- *Created a correlation heatmap with coolwarm colour palette for visual interpretation.*
- *Identified strong, moderate, and weak feature relationships.*

4.4.5 Feature Relationship Analysis

- *Generated pair plots to visualise interactions between all numerical features.*
- *Studied relationships among healthcare-related attributes.*
- *Explored potential factors associated with diabetes outcomes.*

4.4.6 Statistical Evaluation

- *Calculated feature skewness coefficients for all 13 numerical features.*
- *Evaluated distribution characteristics and identified features requiring transformation prior to modelling.*

4.4.7 Documentation Activities

- *Created the notebook 03_EDA_Visualization.ipynb.*
- *Documented observations and analytical findings.*
- *Recorded visualisation outputs and interpretations.*
- *Updated project documentation with EDA results.*

4.5 Summary

Today's work focused on performing Exploratory Data Analysis (EDA) to gain deeper insights into the Diabetes Dataset. Various statistical and visualisation techniques were applied to understand feature distributions, identify trends, detect outliers, and examine relationships between healthcare-related variables. Summary statistics were generated to evaluate the characteristics of numerical features, while histograms and KDE plots were used to analyse data distribution patterns.

Correlation analysis was conducted to understand the relationships among dataset attributes, and a correlation heatmap was created to visualise these interactions effectively. Boxplots were used to identify potential outliers, while pair plots provided a comprehensive view of feature interactions and dependencies. Feature skewness was analysed to better understand the distribution behaviour of numerical variables.

The generated visualisations and analytical findings were documented and stored within the project repository. The notebook `03_EDA_Visualization.ipynb` was completed and synchronised with GitHub. By the end of the session, a comprehensive understanding of the dataset's patterns, distributions, and feature relationships was achieved, providing valuable insights for future analysis, reporting, and decision-making activities.

4.6 Outcomes Achieved

- Successfully explored the Diabetes Dataset through visual analysis.
- Identified feature distributions, trends, and relationships.
- Detected potential outliers and data patterns.
- Generated meaningful visualisations and analytical insights.
- Prepared the dataset and findings for final reporting and review.

5. Insights Generation and Reporting

Source document: Daily Report — 05/06/2026 | Week 2, Day 5

5.1 Overview and Objectives

The objective of Day 5 was to consolidate and interpret the results obtained from previous stages of the Healthcare Facility & Patient Management System project. Findings from dataset understanding, data quality assessment, and exploratory data analysis were reviewed to identify meaningful patterns and healthcare-related insights. The observations were documented in a structured manner to create a comprehensive summary of the work completed during the week. This activity was intended to transform raw analytical results into actionable insights and prepare the project for presentation during the weekly review meeting.

The specific key points addressed were:

- Review and consolidate findings from previous analysis phases.*
- Analyse results obtained from dataset understanding and EDA activities.*
- Identify significant patterns, trends, and relationships within the Diabetes Dataset.*
- Extract meaningful healthcare-related insights from visualisations and statistical analysis.*
- Summarise key observations from data quality assessment and exploratory analysis.*
- Create insight and observation reports for documentation.*
- Organise notebooks, visualisations, and project resources.*
- Prepare project deliverables for the weekly review meeting.*
- Document conclusions and recommendations based on analysis results.*
- Synchronise final updates with the GitHub repository.*

5.2 Source Code — Section 1 (Google Colab)

5.2.1 Step 1 — Library Imports

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
```

Listing 5.1. Library imports (Day 5, Step 1)

5.2.2 Step 2 — Data Loading

```
1 df = pd.read_csv('/content/diabetic_data.csv')
```

Listing 5.2. Dataset loading (Day 5, Step 2)

5.2.3 Step 3 — Dataset Summary for Report

```
1 print("Dataset Shape:", df.shape)
2 print("\nTotal Rows:"      , df.shape[0])
3 print("Total Columns:"    , df.shape[1])
4 print("\nMissing Values:", df.isnull().sum().sum())
5 print("Duplicate Records:", df.duplicated().sum())
```

Listing 5.3. Dataset summary generation for report (Day 5, Step 3)

Output:

```
Dataset Shape: (27793, 50)
```

```
Total Rows      : 27793
```

```
Total Columns : 50
```

```
Missing Values   : 47483
```

```
Duplicate Records: 0
```

Note: At this stage of the project, the dataset used had been further filtered to 27,793 rows (down from 100,227 in the quality assessment session), reflecting additional preprocessing steps applied between sessions.

5.2.4 Step 4 — Summary Table


```
1 summary = pd.DataFrame({
2     'Metric': [
3         'Total Rows',
4         'Total Columns',
5         'Missing Values',
6         'Duplicate Records'
7     ],
8     'Value': [
9         df.shape[0],
10        df.shape[1],
11        df.isnull().sum().sum(),
12        df.duplicated().sum()
13    ]
14 })
15 summary
```

Listing 5.4. Summary metrics DataFrame (Day 5, Step 4)

Table 5.1. Dataset Summary Table Output (Day 5)

P#	PMetric	PValue
0	Total Rows	27,793
1	Total Columns	50
2	Missing Values	47,483
3	Duplicate Records	0

5.2.5 Step 5 — Correlation Analysis

```
1 corr = df.corr(numeric_only=True)
```

Listing 5.5. Correlation matrix computation for reporting (Day 5, Step 5)

5.2.6 Step 6 — Correlation with Outcome (Column Index)

```
1 print(corr.columns)
```

Listing 5.6. Display of correlation column index (Day 5, Step 6)

Output:

```
Index(['encounter_id', 'patient_nbr', 'admission_type_id',  
      'discharge_disposition_id', 'admission_source_id',  
      'time_in_hospital', 'num_lab_procedures', 'num_procedures',  
      'num_medications', 'number_outpatient', 'number_emergency',  
      'number_inpatient', 'number_diagnoses'],  
      dtype='object')
```

5.2.7 Step 7 — Top Feature Relationships

```
1 corr_matrix = corr.abs()  
2 corr_pairs = (  
3     corr_matrix.unstack()  
4     .sort_values(ascending=False)  
5 )  
6 corr_pairs.head(20)
```

Listing 5.7. Top pairwise feature correlations by absolute value (Day 5, Step 7)

Table 5.2. Top 20 Feature Pairs by Absolute Correlation (Day 5)

PFeature 1	PFeature 2	PAbs. Correlation
encounter_id	encounter_id	1.000000
admission_type_id	admission_type_id	1.000000
patient_nbr	patient_nbr	1.000000
number_inpatient	number_inpatient	1.000000
num_procedures	num_procedures	1.000000
num_lab_procedures	num_lab_procedures	1.000000
time_in_hospital	time_in_hospital	1.000000
admission_source_id	admission_source_id	1.000000
discharge_disposition_id	discharge_disposition_id	1.000000
num_medications	num_medications	1.000000
number_outpatient	number_outpatient	1.000000
number_diagnoses	number_diagnoses	1.000000
number_emergency	number_emergency	1.000000
num_procedures	num_medications	0.478829
num_medications	num_procedures	0.478829
num_medications	time_in_hospital	0.477454
time_in_hospital	num_medications	0.477454
admission_type_id	admission_source_id	0.365594
admission_source_id	admission_type_id	0.365594
time_in_hospital	num_lab_procedures	0.323511

5.2.8 Step 8 — Create Insights Table

```
1 insights = pd.DataFrame({
2     'Insight No': [1, 2, 3, 4, 5],
3     'Observation': [
4         'Glucose shows strong relationship with diabetes
5         outcome',
6         'BMI may influence diabetes occurrence',
7         'Outliers detected in multiple features',
8         'Several features show skewed distributions',
```

```
8         'Correlation analysis revealed important feature
          relationships'
9     ]
10 } )
11 insights
```

Listing 5.8. Structured insights DataFrame (Day 5, Step 8)

Table 5.3. Initial Insights DataFrame Output (Day 5)

PInsight	PObservation
No	
1	Glucose shows a strong relationship with diabetes outcome.
2	BMI may influence diabetes occurrence.
3	Outliers detected in multiple features.
4	Several features show skewed distributions.
5	Correlation analysis revealed important feature relationships.

5.2.9 Step 9 — Initial Insights (Narrative Format)

The five initial insights were expanded into narrative observations:

- **Insight 1:** Glucose shows a strong relationship with diabetes outcome.
- **Insight 2:** BMI appears to have moderate influence on diabetes occurrence.
- **Insight 3:** Certain features contain outliers that may affect analysis results.
- **Insight 4:** Several numerical features exhibit skewed distributions, requiring transformation prior to modelling.
- **Insight 5:** Feature relationships identified through correlation analysis can support future predictive modelling.

5.3 Source Code — Section 2 (GitHub Operations)

```
1 git add .
2 git commit -m "Completed Week 1 analysis and reporting"
3 git push
```

Listing 5.9. *Git commit for Week 1 analysis completion (Day 5)*

Repository: <https://github.com/praneeth-tech/HF-PMS.git>

5.4 Work Completed

5.4.1 Insight Generation and Analysis

- Reviewed results from dataset understanding, data quality assessment, and EDA activities.
- Analysed statistical summaries and visualisation outputs.
- Evaluated relationships between healthcare-related features.
- Interpreted findings obtained from correlation analysis.

5.4.2 Pattern and Trend Identification

- Identified significant patterns within the Diabetes Dataset.
- Examined relationships among numerical variables.
- Observed trends revealed through visualisations.
- Evaluated the impact of key healthcare attributes on diabetes-related outcomes.

5.4.3 Correlation Analysis

- Reviewed correlation matrix results.
- Identified strong, moderate, and weak feature relationships.
- Analysed interactions between important healthcare variables.
- Documented significant correlations for future reference.

5.4.4 Insight Documentation

- Generated meaningful observations from analytical findings.
- Created an insights table summarising five major observations.
- Documented trends, patterns, and feature relationships.
- Recorded key findings for project reporting purposes.

5.4.5 Reporting Activities

- Created the notebook `04_Insights_and_Report.ipynb`.

- Prepared summary observations based on completed analyses.
- Developed conclusion statements for the project phase.
- Organised analytical findings into a structured format.

5.4.6 Project Organisation

- Reviewed project files and folder structure.
- Verified availability of datasets, notebooks, and visualisations.
- Organised project resources for the weekly review meeting.
- Ensured proper documentation of completed work.

5.4.7 Version Control Activities

- Added Day 5 notebook and documentation files to Git tracking.
- Created a project commit for insights and reporting activities.
- Successfully pushed updates to the GitHub repository.
- Maintained synchronisation between local and remote repositories.

5.5 Summary

Today's work focused on consolidating and interpreting the results obtained from the previous stages of the Healthcare Facility & Patient Management System project. The findings from dataset understanding, data quality assessment, and exploratory data analysis were reviewed to identify meaningful patterns, trends, and relationships within the Diabetes Dataset. Correlation analysis and visualisation outputs were examined to gain a deeper understanding of healthcare-related variables and their interactions.

Significant observations and insights were documented in a structured manner to transform analytical results into actionable information. An insights table, observation summary, and project conclusions were prepared to capture the most important findings discovered during the analysis process. The notebook `04_Insights_and_Report.ipynb` was created to organise and document these results effectively.

In addition to insight generation, project files, notebooks, visualisations, and documentation were reviewed and organised to ensure completeness and consistency. All updates were synchronised with the GitHub repository using Git version control. By the end of the session,

the analytical findings were successfully consolidated into a comprehensive summary, and the project was fully prepared for the weekly review meeting and future development activities.

5.6 Outcomes Achieved

- *Successfully consolidated findings from all previous analysis phases.*
- *Identified meaningful healthcare-related insights from the dataset.*
- *Created a structured summary of observations and conclusions.*
- *Prepared project deliverables for the weekly review meeting.*
- *Completed Week 1 analytical and reporting activities successfully.*

End of Chapters 1 – 5

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